

Course Title: Wireless Communication and Mobile Computing
Course code:-SWEG 3053

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Target Dept.: Software Engineering
Target Year: III

September 18, 2020

Chapter 1 : Introduction to Wireless Communication

Chapter 2 :Wireless Communication Principle and Fundamentals

Chapter 3 :Wireless Telecommunication Technologies

Chapter 4 :Mobile Computing

Chapter 5 :Security in Wireless Communication

Chapter 6 :Mobile Management

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CHAPTER ONE

Introduction to Wireless Communication

- Basics in Wireless Communication System
- Evolution of Wireless Communication
- Types of Wireless Communication or Network
- Current Wireless Communication System
- Challenges in Wireless Communication
- Standards
- Application areas of wireless networking/communication

Recalling points

- **Data Communication :**
 - A process that allows *information* to pass between a machines (**sender** and **Receiver**). OR
 - Transfer of *meaningful information* between two locations
- Telecommunications
 - Is Type of communication between different parties which are located far apart by a means of electrical or electromagnetic techniques.
- Communication has 5 basic entities :
 - *Transmitter*
 - *Meaningful Information* → **Message**
 - *Transmission Channel or Medium*
 - *Communication Protocol*
 - *Receiver*

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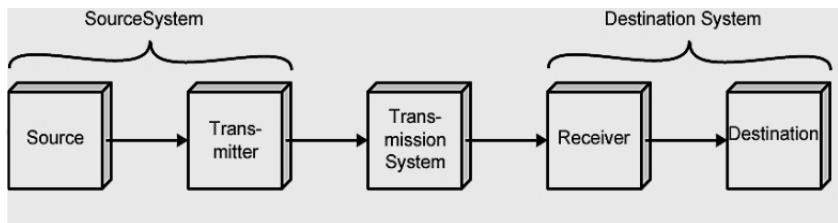
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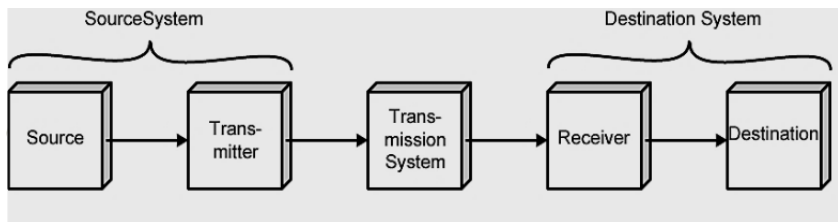
Basics of Wireless Communication System ...



General Block Diagram of Communication

- The effectiveness of such communication is function of :
 - Delivery : → Delivering to the right recipient.
 - Accuracy : → Accurate delivery of the actual data.
 - Timeliness: → Delivering the data at the right time.
 - Jitter : → Uneven delay that may occur during the delivery.

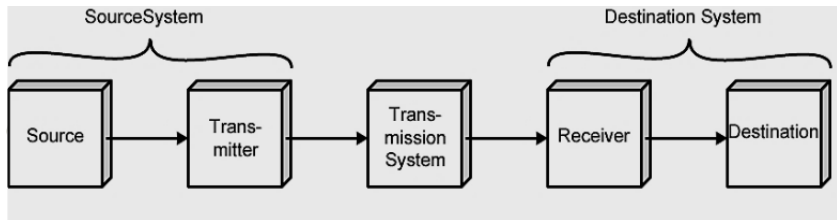
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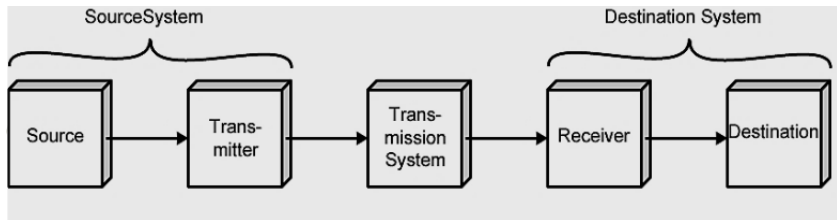
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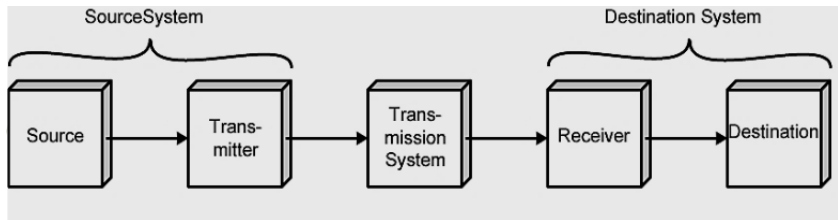
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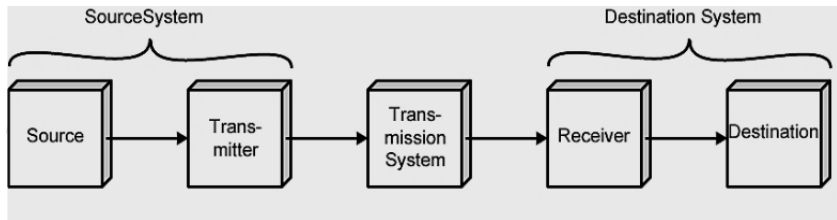
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- **Channel:** is a path between two communication devices
- **Channel capacity:** How much data can be passed through the channel (bit/sec)
 - Also called *Channel bandwidth*
- The smaller the pipe the slower data transfer!
- Channels are Two types:
 - 1 Physical → Wired (Cables)
 - 2 Wireless → Air interface

✓ Wireless Communication Medias

- **Broadcast Radio :**
 - Distribute signals through the air over long distance
 - Uses an antenna
 - Typically for stationary locations

- *Cellular Radio*

- A form of broadcast radio used for *mobile communication*
- High frequency radio waves to transmit *voice or data*
- Utilizes *frequency-reuse*

- *Microwave Channels*

- Radio waves providing high speed transmission
- They are point-to-point (can't be obstructed)
- Used for *satellite communication*

- *Infrared (IR)*

- Wireless transmission media that sends signals using infrared light-waves

✓ What is Wireless Communication ?

✓ is basically transmitting/receiving voice as well as data using EM waves in open space, basically free from wires.

✓ The information from sender → receiver usually carried over *A well defined frequency channel*

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- Wireless communications is one of the fastest growing segments of the telecommunications industry.
- Wireless communication systems, such as *cellular*, *cordless* and *satellite phones* as well as *wireless local area networks (WLANs)* have found widespread use.
- Primitive Wireless Communications systems :
 - Smoke signals
 - Flashing lights
 - Flags
 - etc...

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Evolution of Wireless Networks ...

- Wireless network utilize **Radio Frequency** as Transmission medium.
 - The First Radio transmission Starts in 1895
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- The *first radio-based wireless transmission* between two places → **Guglielmo Marconi**
 - Marconi successfully transmitted a *radio signal* across the *Atlantic Ocean* from *Cornwall* to *Newfoundland*
 - The first bidirectional communication across the Atlantic Ocean was established.
 - Radio-based transmission continued to evolve



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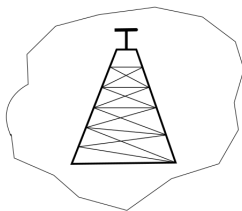
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1 Mobile Telephony System

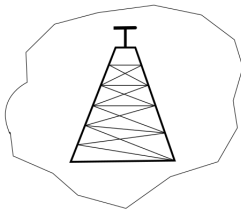
- The first public mobile telephone system → Mobile Telephone System (MTS)
- It was also known as **Car-based mobile Telephony**
 - **Mobile transceivers** of MTS were very big and could be carried only by vehicles.
- MTS was Analog system
 - meaning that it processed voice information as a continuous waveform.
- The system was half-duplex
 - Communication parties @ a time could either speak or listen.
 - To switch between the two modes, users had to push a specific button on the terminal.
- MTS utilized a Base Station (BS) with a single high-power transmitter.



- Base Stations utilized the same frequencies.
- Due to **power limitations**, mobile units transmitted not directly to the BS but to **receiving sites**

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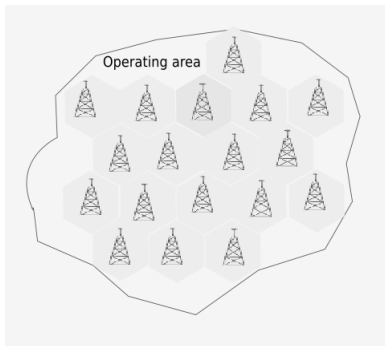
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Mobile Telephon System (MTS)...

- How a call happen in MTS ?
 - The caller first called a special number to connect to an MTS operator.
 - The caller informed the operator of the mobile subscriber's number.
 - Then the operator searched for an idle channel in order to relay the call to the mobile terminal.
- Major limitations:
 - Manually Switching of calls.
 - limited channels are available (3 channels)
- Improved Mobile Telephone System (IMTS)
 - Utilized *automatic call switching*
 - Enhanced its mode of transmission → *Full-duplex*
 - The *intermediation of the operator* in a call Eliminated.
 - Number of channel increased to 23
 - IMTS has **problem**:
 - providing a small capacity Because spectrum usage.
 - Interference to adjacent systems due to *large power of BS transmitter*.

② Analog Cellular Telephony Era

- The first **cellular concept** issued @ AT&T Bell laboratories.
- The cellular concept replaces *high-coverage BSs* with a number of *low-coverage stations*.



- Area of coverage of each such BS → **Cell**
- The system was divided into a set of adjacent, non-overlapping cells.
- The available spectrum is partitioned into **channels**.
- Each BS is connected via wires to a device → **Mobile Switching Center (MSC)**
- The first generation of cellular systems (1G systems) - concept designed
- Analog cellular systems were the first step for the *mobile telephony industry*.

8 Digital Cellular Telephony

- Analog cellular Eras - **Limited performance**



Alleviated in Digital cellular era

- The major issue : *data is represented digitally*
- Voice signals through an Analog to Digital (A/D) converter → bitstream to modulate an RF carrier.
- The major advantage of Digital Era:
 - Privacy and security
 - In digital systems, it is possible to apply **error detection** and **error correction** techniques to the voice bitstream.
 - RF carrier is shared by more than one user, either by using different time slots or different codes per user.
- The Second Generation cellular (2G) network conceived here.

❶ Cordless Phones

- The Invention of Phone has :
 - ❶ Primarily aim to replace the cord of conventional telephones with a wireless link.
 - ❷ To enable user move while speaking → *Mobility*
- Early cordless telephones were analog: → *Poor call Quality*



Changed in 1G of digital cordless telephones

- 1G digital cordless telephone was successful, However *Handset Usage Restriction*
- Telepoint system avoid the restriction → *2G digital cordless phone*
- Problems existed calling is ✓ but not *receiving calls (X)* while on fly
- The other problem is *roaming between telepoint BSs was not supported*

❷ Wireless Data System

- The first wireless data system is developed in 1970's @ University of Hawaii.
- **ALOHANET** Research project

"The idea of the project was to offer bi-directional communications between computers spread over four islands and a central computer on the island of Oahu without the use of phone lines."

- *Wider Area Data System, Wireless Local Area Networks (WLANS), Wireless ATM (WATM), etc*

⑥ Satellite Communication/Network

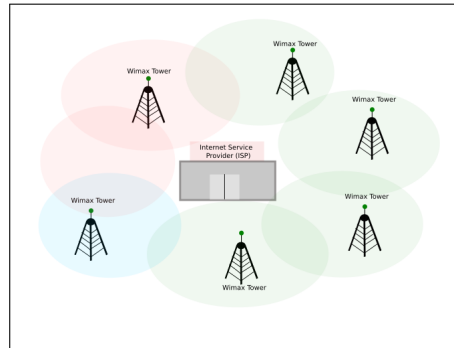
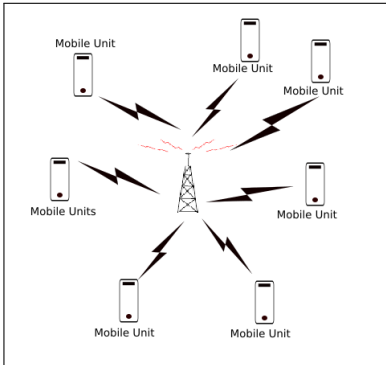
- Most Useful wireless communication technology.
- Utilize the application of Satellite to enable communication regardless of geographical location.
- *What Benefits such communication brought:*
 - No need to bother about the cellphone mast or not.
 - Avoids the need of physical connection.
 - Ensure that users are completely free from telecommunication infrastructure.

Types of Wireless Network

- Wireless Network is categorized in to :

- ① **Infrastructured Wireless Network**

- Mobile Units communicate with each other via fixed infrastructure.
 - Such as **Cellular Communication**, **Wi-Max**



② Infrastructureless Wireless Network

- There is no fixed-inrstructure → Ad-Hoc networks.
- Devices communicate with each other using multi-hop links
- *MANET, VANET, WMNET, WSN, and other.*

• Based on Coverage area Wireless Network are :

① Wireless LANs (WLAN)

- Wi-Fi Networks: IEEE 802.11 (CSMA/CA)
- Ad-hoc Networks: useful when infrastructure not available
- Personal Area Networks (PAN): IEEE 802.15 (Bluetooth)

② Wireless MANs (WMAN)

- WiMAX Networks: IEEE 802.16
- Mesh Networks: Multi-hop Wi-Fi

③ Wireless WANs (WWAN)

- Cellular Networks: GSM, GPRS, UMTS, LTE
- Satellite Networks: Iridium

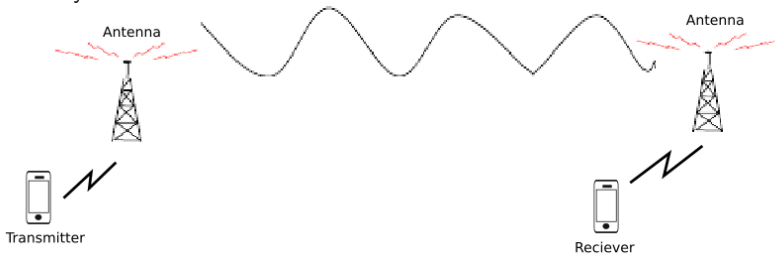
Types of Wireless Communications

Some of the current wireless communication systems:

- 1 Television and Radio Broadcasting
- 2 Satellite Communication
- 3 Radar
- 4 Mobile Telephone System (Cellular Communication)
- 5 Global Positioning System (GPS)
- 6 Infrared Communication
- 7 WLAN (Wi-Fi)
- 8 Bluetooth
- 9 Paging
- 10 Cordless Phones
- 11 Radio Frequency Identification (RFID)

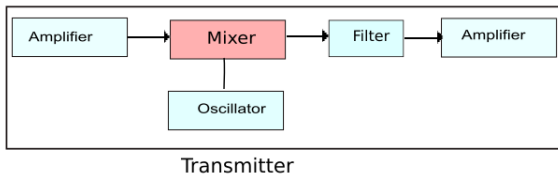
How Wireless transmission ..

- Generally wireless data transmission:



- Wireless Transmission systems consist of:
 - **Transmitters:** → Small devices in Mobile Units for transmitting
 - **Antennas:** radiates electromagnetic energy into air
 - **Receivers:** → Small devices in Mobile Units for Receiving signals.
 - Transmitters + receivers → transceivers

- Every Transmitter/Receiver has different Components



- Suppose you want to generate a signal that is sent at 1000 MHz and the original source generates a signal at 700 MHz.
 - **Amplifier** - strengthens the initial signal
 - **Oscillator** - creates a carrier wave of 300 MHz
 - **Mixer** - combines signal with oscillator and produces 1000 MHz(also does modulation, etc)
 - **Filter** - selects correct frequency
 - **Amplifier** - Strengthens the signal before sending it

Why Wireless Communication ?

- Why Wireless Communication :

- ① Freedom From Wires

- *no cost of installing wires or rewiring* → No bunches of wires running here and there
 - provide **Instantaneous communication** without the need for physical connection setup (Bluetooth, Wi-Fi, Wi-MAX)

- ② Global Coverage

- Communications can reach anywhere
 - in places such as **rural areas, Battlefields, Vehicles, Space(via man-made stars)**

- ③ Stay Connected

- **Roaming Service**: which allows flexibility to stay connected anywhere and anytime

- ④ Flexibility

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- **Multipath Propagation:**

- Signal travels from its source to destination, in between there are obstacles
→ Signal propagate in paths beyond the direct line of sight. why ?
- due to reflections, refraction and diffraction and scattering.

- **Path loss:**

- is the attenuation of the transmitted **signal strength** as it propagates away from the sender.
- Determined as the ratio between *the powers of the transmitted signal* to the *receiver signal*.
- Not the same for every wireless communication.

- **Limited Spectrum:**

- frequency channel/spectrum are shared by many wireless technologies and not by one single wireless technology.

- **Security:**

- is the major concern in any type of communication.

- **Dynamic topology**

- Communication systems that interact with each other require standardization.
- Standards define *what is needed for interoperability*.
- Standards do not include all the details on all aspects of the system design.
- The main goal of standardization is *interoperability*.
- International standardization authorities are :
 - *National Organizations*: established by treaty among national governments,
 - Those comprising voluntary, *nontreaty organizations*.
- Standard Organization for Communication and Networks:
 - ITU: (*International Telecommunication Union*)
 - ETSI
 - ISO: (*International Standardization Organization*)
 - IETF: (*Internet Engineering Task Force*) etc

- IEEE : (Institute of Electrical and Electronics Engineering)
 - "has a standardization group that develops standards in the area of electrical engineering and computing".
 - Families of IEEE :
 - IEEE 802.3 (Ethernet, CSMA/CD)
 - IEEE 802.4 (Token Bus)
 - IEEE 802.5 (Token Ring)
 - IEEE 802.6 (MAN)
 - IEEE 802.11 (WLAN/Wi-Fi, CSMA/CA)
 - IEEE 802.15 (Bluetooth)
 - IEEE 802.16 (WiMAX)

CHAPTER TWO

Wireless Communication Principle and Fundamentals

- The Electromagnetic Spectrum.
- Wireless Communication characteristics and Modeling.
- Analog and Digital Transmission.
- Modulation Techniques of Wireless System
- Multiple access for wireless systems
- Wireless Services: Circuit and Data(packet) model

⇒ What are Electromagnetic waves ?

- were predicted by the British physicist *James Maxwell*
- German physicist *Heinrich Hertz* → Observe
- Electromagnetic waves
 - are created by the movement of electrons and have the ability to propagate through space.
 - are generated via generation of an **electromagnetic field**.



The speed of an electrical charge is changed

- Both transmitters and receiver employ this Emf/Emw
 - *Transmitter* :Vibrates electrons, which are the particles that orbit all atoms and contain electricity.
 - *Receptor* :by examining values of electrical signals that are induced to the receiver's antenna by the incoming wave.
- Electromagnetic waves has 2 fundamental X-stics:
 - *Wave's frequency*:
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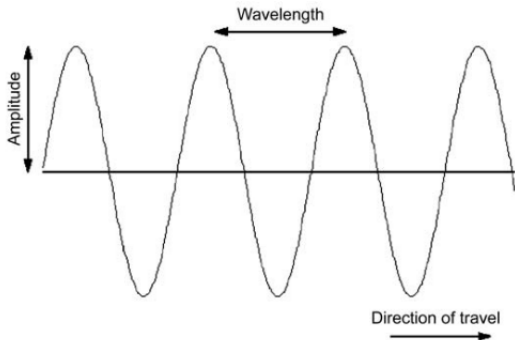


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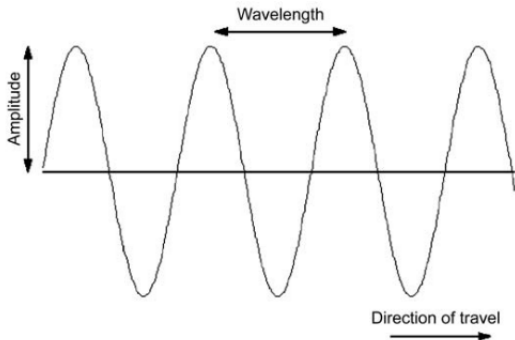
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- **Wavelength** : refers to the distance between two consecutive maximum or minimum peaks of the electromagnetic wave transmission.



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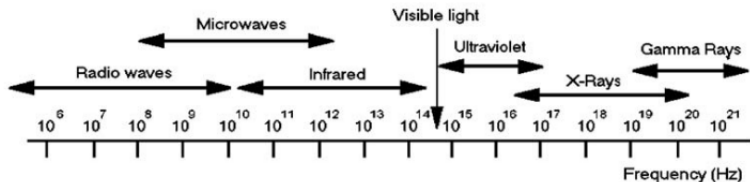
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Electromagnetic Spectrum...

⇒ What are Electromagnetic Spectrum?

- The complete range of electromagnetic radiation.
- The range is divided in to small parts → *bands*



- Frequency is measured on a logarithmic scale → increased by a factor of 10
- *Higher bands* have more bandwidth and can carry more data.

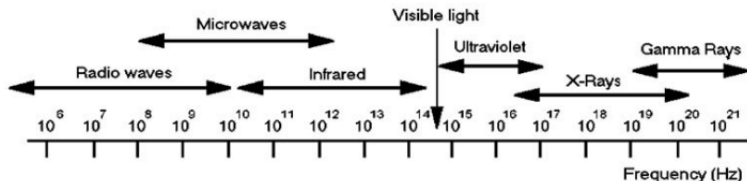


► *not used for wireless transmissions*

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⇒ What are Electromagnetic Spectrum?

- The complete range of electromagnetic radiation.
- The range is divided in to small parts → *bands*



- Frequency is measured on a logarithmic scale → increased by a factor of 10
- *Higher bands* have more bandwidth and can carry more data.

⇓

► *not used for wireless transmissions*

- Why Higher bands not used for wireless transmission?

- Difficult to modulate
- Dangerous for living creatures
- Typically shorter range
- Easily blocked by obstacles.

- ▶ Lower Bands Spectrums

- Radio waves
- Microwaves
- Infrared waves

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- Ultraviolet (UV).
- X-Rays
- Gamma rays

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Electromagnetic Spectrum...

1 Radio Transmission

- Utilize the radio frequency (RF waves)
- The first to be applied for wireless communications.
- RF comes with two flavour :
 - 1 Lower frequency radio bands → TV and radio broadcasting
 - 2 High frequency radio bands → Fast data services + having high bandwidth

2 Microwave Transmission

- They have small wavelengths compared to the other radio waves.
 - High frequency radio bands such as UHF, SHF, EHF
- ↓
- Most Wireless Communication system employ*
- Drawback of microwaves is that *easily attenuated by objects found in their path* ⇒ Multipath fading

3 Infrared Transmission

- Unguided infrared waves are widely used for short-range communication.
- Are relatively directional, cheap, and easy to build.
- Disadvantage *the fact that infrared waves do not pass through solid walls.*

4 Visible light.

- The tiny part of the spectrum between UV and Infrared (IR).

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Wireless communication Characteristics and Modeling

- An important issue in wireless communications is of course the *amount of information that can be carried over a wireless channel*



Bit Rate

- *Shannon's formula* gives the Upper bound on the bit rate W of any channel.

$$W = H \cdot \log_2(1 + S/N)$$

- In real wireless channels the bit rates achieved can be significantly lower why?
 - "Besides thermal noise, there exist a number of impairments on the wireless channels that cause reception errors"
 - Most of these impairments stem from the physics of wave propagation
 - it provides a means for predicting the coverage area of a transmitter and the interference experienced at the receiver.

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1 Free Space Path Loss

- Signal attenuation due to distance between the transmitter and the receiver.
- In free space, the received power is proportional to r^2 , where r is the distance between the transmitter and the receiver.

2 Doppler shift

- is caused when a signal transmitter and receiver are moving relative to one another
- Frequency of the received signal will not be the same as that of the source.
- This phenomenon becomes important when developing mobile radio systems.

3 Reflection

- occurs when an electromagnetic wave falls on an object with dimensions very large compared to the wave's wavelength.

4 Scattering

- when the signal is obstructed by objects with dimensions in the order of the wavelength of the electromagnetic wave.
- This phenomenon causes the energy of the signal to be transmitted over different directions and is the most difficult to predict

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- also known as **shadowing**
- Occurs when an electromagnetic wave falls on an impenetrable object

⇒ Wireless Propagation Modeling

- Predicting the actual signal arriving at a receiver **is too difficult**
- However theoretically, the signal arriving at the receiver needs to be predicted. ⇒ **propagation models**



Defⁿ: "A set of mathematical expressions, algorithms and diagrams that predict the propagation of a signal in a given environment."

- Propagation Model comes in the following flavour:
 - 1 Empirical/Statistical
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⇒ Empirical propagation Modeling

- describe the radio characteristics of an environment based on measurements.
- Implicitly take into account all the factors that affect signal propagation.
- These models are computationally efficient and accurate.



The accuracy is affected by the accuracy of the measurements that are used.

⇒ Theoretical propagation Modeling

- base their predictions not on measurements but on principles of wave theory.
- independent of measurements and thus their predictions are more accurate for a wide range of different environments
- they are expressed by algorithms that are very complex and thus computationally inefficient.

- In terms of the **radio environment they describe**, propagation models can be categorized into **Indoor** and **outdoor models**
- Outdoor subdivided into
 - **Macrocell** models describing propagation over large outdoor areas.
 - **Microcell** models describing propagation over small outdoor areas

⇒ Bit Error Rate (BER) Modeling of Wireless

- A number electromagnetic wave propagation impairments, such as **free-space loss** and **thermal noise, fading**



The primary cause of reception errors in wireless communications

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- **Bit Error Rate** is defined as:

is the degree of a receiving bits with the presence of error.

- Why Bit Error ? Primary reason



- Atmospheric noise
- Physical obstructions found in the signal's path.
- Multipath propagation and Signal interference

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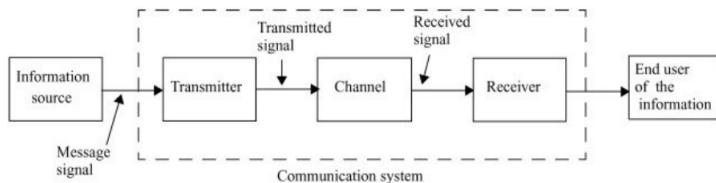
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Analog and Digital Transmission ...



- **Transmittter** : is to change the message signal into a form suitable for transmission over the channel.



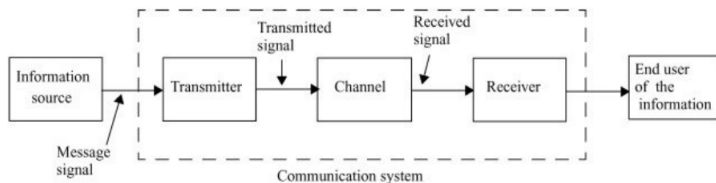
This modification is achieved by means of **Modulation**.

- **Channel** → transmission line
- **Reciever**: for reconstructing the original message signal from the received signal.



This reconstruction is done by means of **Demodulation**.

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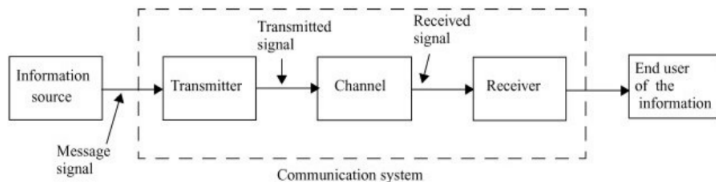
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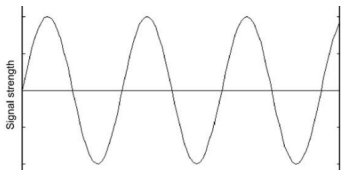
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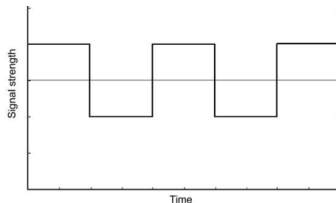
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Analog and Digital Transmission ...

- An important parameter of message relaying between a source and a destination is whether the message is analog signal/data or digital signal/data.



Analog signals take continuous values in time



Digital change between certain levels at specific time positions.

- Analog Transmission

- is a transmission method which convey **information**. using a continious signal differ in Amplitude, phase and other property of waves.



Such as Voice, Image, data, signals, videos and other

- Two possibilities :

- ➊ Analog data via analog transmission



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Radio Transmission

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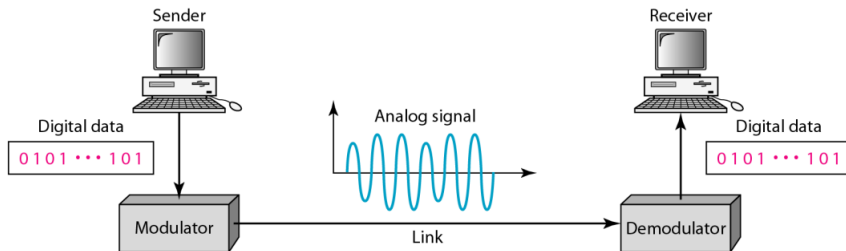


Radio Transmission

- ## 2 Digital data via analog transmission

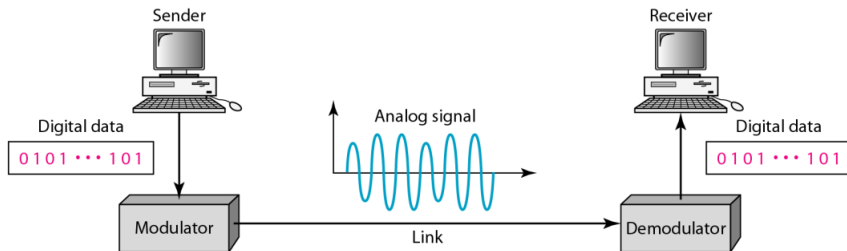


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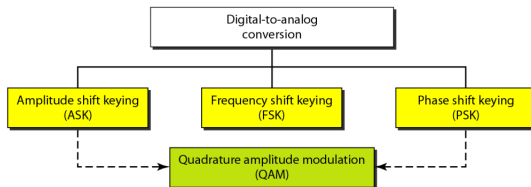
- A carrier signal performs the function of transporting the digital data in an analog waveform.
- The analog carrier signal is manipulated to uniquely identify the digital data being carried.

Analog Transmission ...



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Analog Transmission ...



- Amplitude Shift Keying (ASK): *is implemented by changing the amplitude of a carrier signal to reflect amplitude levels in the digital signal.*
- Frequency Shift Keying: *The digital data stream changes the frequency of the carrier signal, f_c*
- Phase Shift Keying : *by varying the phase shift of the carrier signal to represent digital data.*
- Quadrature phase shift keying (QPSK) : *it actually transmits two bits per symbol. In other words, a QPSK symbol doesn't represent 0 or 1—it represents 00, 01, 10, or 11.*

- What is Multiple Access ?

- **Multiple access** method allows several terminals connected to the same multi-point transmission medium to transmit over it.
- Multiple access schemes to allow many users to share simultaneously a finite amount of radio spectrum resources.

- There are two channel favours :

- ① Control Channel :

- Forward (downlink) control channel
 - Reverse (uplink) control channel

- ② Traffic channel

- Forward traffic (information) channel
 - Reverse traffic (information) channel

Multiple access of wireless Communication systems

- Multiple Access Techniques

- ➊ Frequency division multiple access (FDMA)
- ➋ Time division multiple access (TDMA)
- ➌ Code division multiple access (CDMA)
- ➍ Space division multiple access (SDMA)

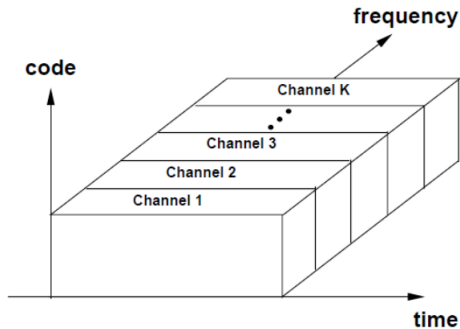
- Frequency division multiple access (FDMA):

- The frequency band is divided into **channels of equal bandwidth** so that each conversation is carried on a different frequency.
- A specific frequency band is given to one person, and it will be received by identifying each of the frequencies on the receiving end.
- It is often used in the first generation of analog mobile phones.
- used in analog Advanced Mobile Phone System (AMPS) and D-AMPS

Frequency division multiple access (FDMA)

- is a method that lets more than one user share a single radio frequency spectrum.
- This is done by assigning active users different frequencies.
- It puts each call in a separate frequency band

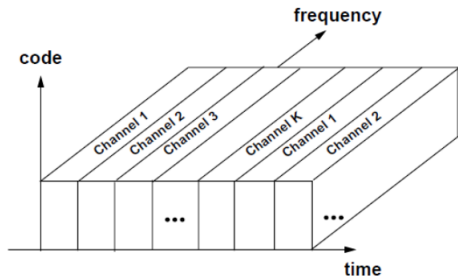
Multiple access of wireless Communication systems...



Time division multiple access (TDMA)

- Stands for **Time division Multiple Access**
- Time frame is divided into slots (channels). Each user is allocated a particular time slot or channel.
- Time Division Multiple Access (TDMA) is a digital cellular telephone communication technology.
- It facilitates many users to share the same frequency without interference.
- Its technology divides a signal into different timeslots, and increases the data carrying capacity.
- In TDMA system signaling dimensions are divided along the Time axis into non overlapping channels and each end users assigned a different timeslot channel.

Multiple access of wireless Communication systems

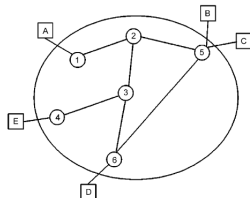


Assignment Two : Code division multiple access (CDMA)Vs Space division multiple access(SDMA)

- Differentiate The two multiple access techniques
- Their Application in telecom technologies
-

- In all communication systems, including wireless systems, transmission of information, (i.e voice or data-related) between a source and a destination station not direct.
- The intermediate nodes are also referred to as switching nodes and the network is known as a switched network.
- Two type of switched ntk :
 - ① Circuit Switched
 - ② Packet Switched
- Circuit switched
 - There is a deicated link between the Communicating parties;
 - For a data transfer to take place in a circuit switched network, → 3 Procedures

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- Circuit establishment

- A dedicated sequence of links between nodes that connect the source to the destination must be defined.

- How circuit is established between A and B.

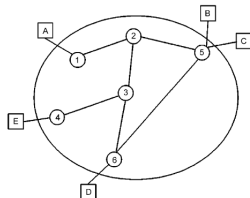
- 1 A send a request for circuit establishment to node 1.
- 2 1 knows B is attached to node 5, finds a way to 5
- 3 Will find in Routing table node 2
- 4
- 5 $A \rightarrow 1 \rightarrow 2 \rightarrow 5 \rightarrow B$

- Transfer of data

- That entails the transfer of the data between the communicating nodes.

- Circuit release

- The circuit is released
- De-allocation of the corresponding resources in the intermediate nodes that serve the circuit.



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What are the pro Ckt Stch

- It is suitable for long continuous transmission, since a continuous transmission route is established, that remains throughout the conversation.
- The dedicated path ensures a steady data rate of communication.
- No intermediate delays are found once the circuit is established.
- Which results with a suitable real time communication of both voice and data transmission

What are the Cons pkt Stch

- The dedicated connection cannot be used for transmitting any other data, even if the data load is very low.
- Bandwidth requirement is high even in cases of low data volume.
- There is underutilization of system resources.
- Once resources are allocated to a particular connection, they cannot be used for other connections.
- Time required to establish connection may be high.

Packet switching

- Packet switching works by transmitting **packets** which most of the times are relatively small.
- Large packets broken into a number of smaller packets.
- Apart from the user's data, each packet carries a control header, which contains information that the network needs to deliver the packet to its destination.
- packet can follow a different route inside the switched network in order to reach its destination.
- In each switching node, incoming packets are stored and the node has to pick up one of its neighbors to hand it the packet.

Questions?!